US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250280898
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	YE; XIAOCUI

DOUBLE-LAYER HANDLEBAR BELT WITH WIDE AND NARROW PORTIONS, AND METHOD AND EQUIPMENT FOR WEAVING THE DOUBLE-LAYER HANDLEBAR BELT WITH WIDE AND NARROW PORTIONS

Abstract

The present disclosure provides a double-layer handlebar belt with wide and narrow portions, a method and an equipment for weaving the double-layer handlebar belt with wide and narrow portions. The double-layer handlebar belt includes: single-layer segments; first double-layer segments alternately connected with the single-layer segments, the first double-layer segment includes a first layer and a second layer, the first layer is arranged below the second layer, a width of the first layer is greater than that of the second layer; second double-layer segments alternately connected with the single-layer segments, the second double-layer segment includes a third layer and a fourth layer, the third layer is arranged below the fourth layer, a width of the third layer is greater than that of the fourth layer, and the fourth layer has a hollow structure, the single-layer segment, the first double-layer segment and the second double-layer segment are woven in one piece.

Inventors: **YE; XIAOCUI (DONGGUAN, CN)**

Applicant: **DONGGUAN POLYUNIONTEXTILE TECHNOLOGY CO.LTD** (DONGGUAN, CN)

Family ID: **1000008494545**

Appl. No.: **19/068222**

Filed: **March 03, 2025**

Foreign Application Priority Data

CN	202410259466.7	Mar. 07, 2024

Publication Classification

Int. Cl.: **A41B9/00** (20060101)

U.S. Cl.:

CPC **A41B9/001** (20130101);

Background/Summary

FIELD
[0001] The present disclosure relates to the field of weaving, especially relates to a double-layer handlebar belt with wide and narrow portions, a method and an equipment for weaving the double-layer handlebar belt with wide and narrow portions.
BACKGROUND
[0002] Woven belts are widely used in the textile industry and other daily life, especially in the apparel industry. Woven belt is an indispensable accessory in the lingerie industry.
[0003] The related technology has already been used to manufacture different types of Woven belts. But the double-layer webbing woven by the related weaving technology cannot provide a good using experience.
SUMMARY
[0004] The present disclosure provides a double-layer handlebar belt with wide and narrow portions, and a method and an equipment for weaving the double-layer handlebar belt with wide and narrow portions, for reducing the weight of the belt and the production cost.
[0005] In one aspect, the present disclosure provides a double-layer handlebar belt with wide and narrow portions, which includes: single-layer segments; first double-layer segments alternately connected with the single-layer segments, the first double-layer segment includes a first layer and a second layer, the first layer is arranged below the second layer, a width of the first layer is greater than that of the second layer; and second double-layer segments alternately connected with the single-layer segments, the second double-layer segment includes a third layer and a fourth layer, the third layer is arranged below the fourth layer, a width of the third layer is greater than that of the fourth layer, and the fourth layer has a hollow structure, wherein the single-layer segments, the first double-layer segments and the second double-layer segment are woven in one piece.
[0006] Furthermore, the single-layer segment comprises a first single-layer segment and a second single-layer segment; a first end of the first layer and a first end of the second layer are connected to the first single-layer segment; a second end of the first layer and a second end of the second layer are connected with a first end of the second single-layer segment; and a first end of the third layer and a first end of the fourth layer are connected to a second end of the second single-layer segment.
[0007] Furthermore, the second layer further includes a third end arranged at a middle of the second layer; the first end and the second end of the second layer are respectively arranged at two ends of the second layer; wherein a width of the first end of the second layer and a width of the second end of the second layer are greater than a width of the third end of the second layer.
[0008] Furthermore, the second layer is arc shaped with respect to the first layer.
[0009] Furthermore, the single-layer segment, the first double-layer segment and the second double-layer segment are made of elastic fabrics or non-elastic fabrics.
[0010] Furthermore, a material shrinkage rate of the first layer is greater than that of the second layer, and a material shrinkage rate of the third layer is greater than that of the fourth layer.

[0011] In another aspect, the present disclosure provides a method for weaving a double-layer handlebar belt with wide and narrow portions, which includes: weaving single layer segments; weaving first double layer segments, the first double-layer segment includes a first layer and a second layer, the first layer is arranged below the second layer, a width of the first layer is greater than that of the second layer; weaving second double layer segments, the second double-layer segment includes a third layer and a fourth layer, the third layer is arranged below the fourth layer, a width of the third layer is greater than that of the fourth layer, and the fourth layer has a hollow structure, wherein the single layer segments are connected with the double-layer segments alternately, and the single layer segments and the double-layer segments are woven in one piece.
[0012] Furthermore, the operation "weaving first double layer segments" includes: weaving a first layer and a second layer respectively; wherein when weaving the second layer, controlling a steel buckle to move upwards, and the woven yarn moving upwards based on the steel buckle from a first position of the steel buckle to a second position of the steel buckle, and reducing a conveying capacity of the weft yarn, as such the width of the second layer is reduced from a first width to a second width; controlling the steel buckle to move downwards, the woven yarn moving downwards based on the steel buckle from the second position of the steel buckle to the first position of the steel buckle, and increasing the conveying capacity of the weft yarn, as such the width of the second layer is increased from the second width to the first width; and a width of the first position of the steel buckle is greater than that of the second position of the steel buckle.
[0013] Furthermore, the operation "weaving second double layer segments" includes: weaving a third layer and a fourth layer respectively; when weaving the fourth layer, controlling a steel buckle to move upwards, and the woven yarn moving upwards based on the steel buckle from a first position of the steel buckle to a second position of the steel buckle, and reducing a conveying capacity of the weft yarn, as such the width of the second layer is reduced from a first width to a second width; wherein a width of the first position of the steel buckle is greater than that of the second position of the steel buckle.
[0014] Furthermore, the single-layer segment includes a first single-layer segment and a second single-layer segment; a first end of the first layer and a first end of the second layer are connected to the first single-layer segment; a second end of the first layer and a second end of the second layer are connected with a first end of the second single-layer segment; and a first end of the third layer and a first end of the fourth layer are connected to a second end of the second single-layer segment.
[0015] Furthermore, the second layer further includes a third end arranged at a middle of the second layer; the first end and the second end of the second layer are respectively arranged at two ends of the second layer; wherein a width of the first end of the second layer and a width of the second end of the second layer are greater than a width of the third end of the second layer.
[0016] Furthermore, the second layer is arc shaped with respect to the first layer.
[0017] Furthermore, the single-layer segment, the first double-layer segment and the second double-layer segment are made of elastic fabrics or non-elastic fabrics.
[0018] Furthermore, a material shrinkage rate of the first layer is greater than that of the second layer, and a material shrinkage rate of the third layer is greater than that of the fourth layer.
[0019] In a further aspect, the present disclosure provides an equipment for weaving a double-layer handlebar belt with wide and narrow portions, wherein the equipment is configured to execute the above method and weave the above double-layer handlebar belt with wide and narrow portions.
[0020] Furthermore, the operation "weaving first double layer segments" executed by the equipment includes: weaving a first layer and a second layer respectively; wherein when weaving the second layer, controlling a steel buckle to move upwards, and the woven yarn moving upwards based on the steel buckle from a first position of the steel buckle to a second position of the steel buckle, and reducing a conveying capacity of the weft yarn, as such the width of the second layer is reduced from a first width to a second width; controlling the steel buckle to move downwards, the

woven yarn moving downwards based on the steel buckle from the second position of the steel buckle to the first position of the steel buckle, and increasing the conveying capacity of the weft yarn, as such the width of the second layer is increased from the second width to the first width; and a width of the first position of the steel buckle is greater than that of the second position of the steel buckle.
[0021] Furthermore, the operation “weaving second double layer segments” executed by the equipment includes: weaving a third layer and a fourth layer respectively; when weaving the fourth layer, controlling a steel buckle to move upwards, and the woven yarn moving upwards based on the steel buckle from a first position of the steel buckle to a second position of the steel buckle, and reducing a conveying capacity of the weft yarn, as such the width of the second layer is reduced from a first width to a second width; wherein a width of the first position of the steel buckle is greater than that of the second position of the steel buckle.
[0022] Furthermore, the single-layer segment includes a first single-layer segment and a second single-layer segment; a first end of the first layer and a first end of the second layer are connected to the first single-layer segment; a second end of the first layer and a second end of the second layer are connected with a first end of the second single-layer segment; and a first end of the third layer and a first end of the fourth layer are connected to a second end of the second single-layer segment.
[0023] Furthermore, the second layer further includes a third end arranged at a middle of the second layer; the first end and the second end of the second layer are respectively arranged at two ends of the second layer; wherein a width of the first end of the second layer and a width of the second end of the second layer are greater than a width of the third end of the second layer.
[0024] Furthermore, the second layer is arc shaped with respect to the first layer.
[0025] The beneficial effects of the present disclosure include: different from the prior art, the double-layer handlebar belt with wide and narrow portions of the present disclosure includes: single-layer segments; first double-layer segments alternately connected with the single-layer segments, the first double-layer segment includes a first layer and a second layer, the first layer is arranged below the second layer, and a width of the first layer is greater than a width of the second layer; second double-layer segments alternately connected with the single-layer segments, the second double-layer segment includes a third layer and a fourth layer, the third layer is arranged below the fourth layer, and a width of the third layer is greater than a width of the fourth layer; the fourth layer is hollow; the single-layer segments, the first double-layer segments and the second double-layer segments are woven in one piece. Through the above-mentioned mode, the first double-layer segment has two layers with different widths, the production cost can be reduced, the weight of the finished product is reduced due to the width reduction, and the fourth layer of the second double-layer segment which can be used as a handle is hollow, so the handlebar belt is much more lightweight and comfortable when used, and the user experience is improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0026] Implementations of the present disclosure will now be described, by way of embodiment, with reference to the attached figures. It should be understood, the drawings are shown for illustrative purpose only, for ordinary person skilled in the art, other drawings obtained from these drawings without paying creative labor by an ordinary person skilled in the art should be within scope of the present disclosure.
[0027] FIG. **1** is a structure view of a double-layer handlebar belt with wide and narrow portions according to an embodiment of the present disclosure.
[0028] FIG. **2** is a top view of the double-layer handlebar belt of FIG. **1**.

[0029] FIG. **3** is a structure view of a second layer C as shown in FIG. **1**.
[0030] FIG. **4** is a structure view of a color selection system selecting a weft yarn of the present disclosure.
[0031] FIG. **5** is a structure view of a weft needle of the present disclosure.
[0032] FIG. **6** is a partial structure diagram of weaving a single-layer segment of the present disclosure.
[0033] FIG. **7** is a partial structure diagram of weaving a double-layer segment of the present disclosure.
[0034] FIG. **8** is a structure diagram of a steel buckle of the present disclosure.
[0035] FIG. **9** is a flow chart of a method for weaving a double-layer handlebar belt with wide and narrow portions according to an embodiment of the present disclosure.
[0036] FIG. **10** is a structure view of a double-layer handlebar belt with wide and narrow portions according to another embodiment of the present disclosure.

DETAILED DESCRIPTION

[0037] Technical solutions in the embodiments of the present disclosure will be clearly and completely described below by referring to the accompanying drawings in the embodiments of the present disclosure. Obviously, the described embodiments are only a part of, but not all of, the embodiments of the present disclosure. All other embodiments obtained by a person of ordinary skill in the art based on the embodiments in the present disclosure and without making creative work shall fall within the scope of the present disclosure.
[0038] The “embodiments” herein means that particular features, structures or characteristics described in conjunction with an embodiment may be included in at least one embodiment of the present disclosure. This phrase at various points in the specification does not necessarily refer to the same embodiment, nor is it a separate or alternative embodiment that is mutually exclusive with other embodiments. It is understood explicitly and implicitly by those skilled in the art that embodiments described herein may be combined with other embodiments.
[0039] Referring to FIG. **1**, FIG. **1** is a structure diagram of a double-layer handlebar belt with wide and narrow portions according to a first embodiment of the present disclosure, and the woven belt includes single-layer segments **1** and first double-layer segments **2**.
[0040] In some embodiments, the double-layer handlebar belt with wide and narrow portions includes a plurality of single-layer segments **1**, a plurality of first double-layer segments **2** and at least one second double-layer segment **3**. As shown in FIG. **1**, the first double-layer segments **2** are alternately connected with the single-layer segments **1**, one single-layer segment **1** is connected between each two first double-layer segments **2**, and one first double-layer segment **2** is connected between each two single-layer segments **1**. One second double-layer segment **3** is connected between each two single-layer segments **1**.
[0041] The first double-layer segment **2** includes a first layer B and a second layer C, and the first layer B is arranged below the second layer C. A width of the first layer B is greater than that of the second layer C.
[0042] The second double-layer segment **3** includes a third layer E and a fourth layer F, and the third layer E is arranged below the fourth layer F. A width of the third layer E is greater than that of the fourth layer F. The fourth layer F has a hollow structure. The hollow structure is filled with yarn or other items to increase user's holding experience.
[0043] As shown in FIG. **1**, the second layer C is arc-shaped with respect to the first layer B, forming an arch. The fourth layer F is arc-shaped with respect to the third layer E, forming an arch. The single-layer segment **1**, the first double-layer segments **2** and the second double-layer segment **3** are made of elastic fabrics or non-elastic fabrics.
[0044] The elastic fabrics mainly refer to knitted fabrics, or product woven from a first fiber selected at least one from a group consisting of nylon, lycra, modal and the like, or product woven from the first fiber blended with a second fiber selected at least one from a group consisting of

cotton, linen, silk, wool and the like.

[0045] Referring to FIG. **2**, FIG. **2** is a top view of FIG. **1**. A width L**1** of the first layer B is greater than a width L**2** of the second layer C; a width L**1** of the third layer E is greater than a width L**3** of the fourth layer F. The single-layer segments **1**, the first double-layer segments **2** and the second double-layer segment **3** are woven in one piece.

[0046] In some embodiments, the single-layer segment **1** includes a first single-layer segment A and a second single-layer segment D; the first single-layer segment A and the second single-layer segment D are arranged at two ends of the first double-layer segment **2** respectively, and a first end of the first layer B and a first end of the second layer C are connected to the first single-layer segment A; a second end of the first layer B and a second end of the second layer C are connected with a first end of the second single-layer segment D. A first end of the third layer E and a first end of the fourth layer F are connected to a second end of the second single-layer segment D.

[0047] Referring to FIG. **3**, the second layer C includes a first end C**1**, a second end C**2**, and a third end C**3**. The first end C**1** and the second end C**2** of the second layer C are respectively arranged at two ends of the second layer C, and the third end C**3** is located at the middle of the second layer C. The widths of the first end C**1** of the second layer C and the second end C**2** of the second layer C are greater than the width of the third end of the second layer C. So that the second layer C presents a wide-narrow-wide effect from the first end C**1**, to the third end C**3** and then to the second end C**2**.

[0048] A material shrinkage rate of the first layer B is greater than that of the second layer C. So that a length of the first layer B is reduced after weaving the double-layer segment, and then the second layer C presents an arc shape.

[0049] The material shrinkage rate of the first layer B is greater than that of the second layer C. So that the length of the first layer B is reduced after weaving the double-layer segment, and the second layer C presents an arc shape.

[0050] Further, in conjunction with FIG. **4**, FIG. **5**, FIG. **6**, FIG. **7** and FIG. **8**, the double-layer handlebar belt with wide and narrow portions of the present disclosure is described:

[0051] The double-layer handlebar belt with wide and narrow portions as shown in FIG. **1** is formed by interweaving the warp and weft yarns using two weft needles and two sheds.

[0052] Specifically, a spindle of the weaving equipment rotates and brings a steel buckle linkage to perform an eccentric movement, as such a steel buckle seat swings back and forth. When swings back and forth, the steel buckle seat brings a weft aluminum hand to swing in an arc, the weft needles are fixed on the weft aluminum hand, and the weft needles are hooked into the knitting needle with the weft yarn after passing through the sheds through the swing of the weft aluminum hand. Two weft needles pass through the upper and lower sheds, respectively.

[0053] The sheds are opening formed by the up and down movements of the warp yarn, including an upper shed and a lower shed, the upper shed and the lower shed are formed under the drive of a brown frame, an upper shed brown frame stroke is from top to middle, and a lower shed brown frame stroke is from middle to bottom. The upper shed is formed by the up and down movements of the brown frame, and the lower shed is formed by the up and down movements of the brown frame, and the two weft yarns pass through the two sheds respectively, and are interwoven into the double-layer handlebar belt with wide and narrow portions as shown in FIG. **1** by the opening and closing of the sheds.

[0054] In some embodiments, the weaving equipment includes fourteen brown frames which are numbered 1 to 14, respectively. The upper shed is formed by the up and down movements of the second to sixth brown frames, and the lower shed is formed by the up and down movements of the seventh to fourteenth brown frames.

[0055] Specifically, when weaving the single-layer segment **1**, all warp yarns are driven by the brown frame to convert from the whole-upper and the middle-upper of the upper structure to the middle-lower and the whole-lower of the lower structure to form the lower shed, and the special Y-shaped steel buckle is positioned between 1 and 2 to weave the lower layer of the single layer segment **1**. In some embodiments, as shown in FIG. **4**, the first brown frame drives a weft yarn color selection system **20** to connect with the first weft yarn **40**, and the second weft yarn is separated from a groove in the first weft needle when the first brown frame moves downwards, and the second weft yarn does not participate in weaving. The groove is groove F as shown in FIG. **5**. The second weft needle **30** is woven in conjunction with the first weft yarn **40** to form the single-layer segment **1**. As shown in FIG. **6**, the second weft needle **30** and the first weft yarn **40** form the single layer segment **1** with the warp yarn when move from middle to bottom driven by the brown frame in the lower shed G. It should be understood that because two weft yarns are needed to weave the first double-layer segment **2**, but only one weft yarn is needed when weaving the single-layer segment **1** weaving, the weft yarn color selection system is required to select a target weft yarn to participate in the weaving of single-layer segment **1**.

[0056] When weaving the first double-layer segment **2**, a part of the warp yarn is transformed from the middle to bottom movements to the middle to top movements under the driven of the brown frame, to form the upper shed. When the first brown frame moves to the top, the weft yarn hooks into the groove in the first weft needle, to participate in weaving. At this time, the first and second weft needles are woven simultaneously to form the first double-layer segment **2**. As shown in FIG. **7**, the second weft needle **30** and the first weft yarn **40** form the first layer B with the warp yarn in the lower shed G, and the first weft needle **50** and the second weft yarn **60** form the second layer C with the warp yarn in the upper shed H.

[0057] Specifically, the present disclosure is illustrated combined with FIG. **8**: when weaving the first double-layer segment **2**, the steel buckle raises from a low position, a fixing shaft of the steel buckle moves upwards and the steel buckle also moves upwards, and the warp yarn passes through a lower half of the steel buckle which is narrower, as such a warp yarn density changes from small to large and less weft yarn is delivered. So that, the woven belt changes from wide to narrow, forming the shape of C**1** to C**3** as shown in FIG. **3**.

[0058] The steel buckle moves from the top position to the lower position, the fixing shaft of the steel buckle moves downwards, the steel buckle also moves downward, the yarn passes through an upper part of the steel buckle, and the upper part of the steel buckle is wider, as such the warp yarn density changes from large to small and the weft yarn is delivered faster, so the woven belt changes from narrow to wide, forming the shape of C**3** to C**2** as shown in FIG. **3**.

[0059] For example, when weaving the single layer segment **1**, the shed in the steel buckle moves from steel buckle position **2** to steel buckle position **4**.

[0060] When weaving the second layer C of the first double-layer segment **2**, the shed in the steel buckle moves from steel buckle position **1** to steel buckle position **3**, and gradually the shed in the steel buckle moves from the steel buckle position **2** to steel buckle position **3**, forming the shape of C**1** to C**3** as shown in FIG. **3**, then the shed in the steel buckle moves from steel buckle position **1** to steel buckle position **3**, forming the shape of C**3** to C**2** as shown in FIG. **3**.

[0061] When weaving the first layer B, the shed in the steel buckle moves from steel buckle position **2** to steel buckle position **4**.

[0062] The structure and material of the first layer are different from those of the second layer, the material shrinkage rate of the first layer is greater than that of the second layer, and the first layer is shorter than the second layer due to the structure and material shrinkage after the double layer is weaved, so the second layer is arc-shaped relative to the first layer.

[0063] When weaving the second double-layer segment **3**, the steel buckle is controlled to move upwards, and the woven yarn moves upwards based on the steel buckle from the first position of the steel buckle to the second position of the steel buckle, and a conveying capacity of the weft yarn is reduced, as such the width of the second layer reduces from the first width to the second width until the weaving is completed. The width of the first position of the steel buckle is greater than that of the second position of the steel buckle.

[0064] Different from the prior art, the present disclosure discloses a double-layer handlebar belt with wide and narrow portions which includes: single-layer segments; first double-layer segments alternately connected with the single-layer segments, and the first double-layer segment includes a first layer and a second layer, the first layer is arranged below the second layer, and a width of the first layer is greater than that of the second layer; second double-layer segments alternately connected with single-layer segments, the second double-layer segment includes a third layer and a fourth layer, the third layer is arranged below the fourth layer, and a width of the third layer is greater than that of the fourth layer; the fourth layer has a hollow structure; the single-layer segment, the first double-layer segment and the second double-layer segment are woven in one piece. Through the above-mentioned mode, the first double-layer segment has two layers with different widths, the production cost can be reduced, the weight of the finished product is reduced due to the reduction of width, and the fourth layer of the second double-layer segment with a hollow structure can be used as a handlebar, which is much more lightweight and comfortable when used, and the user experience is improved.

[0065] Referring to FIG. **9**, FIG. **9** is a flow chart of the method for weaving the double-layer handlebar belt with wide and narrow portions according to a first embodiment of the present disclosure, which includes: [0066] Step **121**: Weaving single layer segments. [0067] Step **122**: Weaving first double layer segments.

[0068] The double-layer segment includes a first layer and a second layer, the first layer is arranged below the second layer, and a width of the first layer is greater than that of the second layer.

[0069] Step **122** can be used to weave the first layer and second layer, respectively.

[0070] When weaving the second layer, the steel buckle is controlled to move upwards, and the woven yarn moves upwards based on the steel buckle from the first position of the steel buckle to the second position of the steel buckle, and the conveying capacity of the weft yarn is reduced, as such the width of the second layer is reduced from the first width to the second width; the steel buckle is controlled to move downwards, the woven yarn moves downwards based on the steel buckle from the second position of the steel buckle to the first position of the steel buckle, and the conveying capacity of the weft yarn is increased, as such the width of the second layer is increased from the second width to the first width; and the width of the first position of the steel buckle is greater than that of the second position of the steel buckle.

[0071] The present disclosure is illustrated combined with FIG. **8**: when weaving the double-layer segment, the steel buckle rises from a lower position, a fixing shaft of the steel buckle moves upwards, and the steel buckle also moves upwards, and the warp yarn passes through a lower half of the steel buckle which is narrower, as such a warp yarn density changes from small to large and less weft yarn is delivered. So that, the woven belt changes from wide to narrow, forming the shape of C**1** to C**3** as shown in FIG. **3**. The steel buckle moves from the top position to the lower position, the fixing shaft of the steel buckle moves downwards, the steel buckle also moves downwards, the yarn passes through an upper part of the steel buckle, and the upper part of the steel buckle is wider, as such the warp yarn density changes from large to small and the weft yarn is delivered faster, so the woven belt changes from narrow to wide, forming the shape of C**3** to C**2** as shown in FIG. **3**.

[0072] For example, when weaving the single layer segment, the shed in the steel buckle moves from steel buckle position **2** to steel buckle position **4**.

[0073] When weaving the second layer C of the first double-layer segment **2**, the shed in the steel buckle moves from steel buckle position **1** to steel buckle position **3**, and the shed in the steel buckle gradually moves from the steel buckle position **2** to steel buckle position **3**, forming the shape of C**1** to C**3** as shown in FIG. **3**, then the shed in the steel buckle moves from steel buckle position **1** to steel buckle position **3**, forming the shape of C**3** to C**2** as shown in FIG. **3**.

[0074] When weaving the first layer B, the shed in the steel buckle moves from steel buckle position **2** to steel buckle position **4**.

[0075] Step **123**: Weaving a second double layer.

[0076] Weaving a third layer and a fourth layer, respectively.
[0077] When weaving the fourth layer, the steel buckle is controlled to move upwards, and the woven yarn moves upwards based on the steel buckle from the first position of the steel buckle to the second position of the steel buckle, and a conveying capacity of the weft yarn is reduced, as such the width of the second layer reduces from the first width to the second width. The width of the first position of the steel buckle is greater than that of the second position of the steel buckle.
[0078] The single-layer segments are woven alternately with the first double-layer segments and the second double-layer segment, and the single-layer segments are woven with the first double-layer segments and the second double-layer segment in one piece.
[0079] In some embodiments, the method for weaving the double-layer handlebar belt with wide and narrow portions includes:
[0080] The structure includes whole-upper, middle-upper, middle-lower and whole-lower which form two sheds, and the lifting of the special Y-shaped steel buckle as shown in FIG. **8** is also adopted. The implementation method is completed by alternately connecting the single-layer segments **1**, the first double-layer segments **2** and the second double-layer segment **3**.
[0081] When weaving the single-layer segment **1**, the whole-upper and the middle-upper of the upper structure is changed to the middle-lower and the whole-lower of the lower structure to form the lower shed, and the special Y-shaped steel buckle is positioned between position **1** and position **2** to weave the lower layer of the single layer segment **1**. The steel buckle position **2** corresponds to the first position described above.
[0082] When weaving the first double-layer segment **2**, the upper layer structure includes whole-upper and middle-upper, the lower layer structure includes middle-lower and whole-lower, and two sheds are formed, and the special Y-shaped steel buckle rises and falls back and forth between position **2** and position **3** to weave the double-layer segment having an upper layer with a change in width and a lower layer with no change in width. The buckle position **3** corresponds to the second position described above.
[0083] When weaving the second double-layer segment **3**, two sheds are used, the upper layer is hollow, and the special Y-shaped steel buckle rises from position **2** to position **3**, and the double-layer handlebar belt is woven with the upper layer handle (the fourth layer) and the lower layer (the third layer) with no change in width, and the hollow portion is filled with yarn. The shrinkage rate of the lower layer is greater than that of the upper layer, the upper layer has an arc shape. The double-layer handlebar belt with wide and narrow portions is woven as shown in FIG. **10**.
[0084] Different from the prior art, the method for weaving the double-layer handlebar belt with wide and narrow portions of the present disclosure can weave the double-layer handlebar belt with wide and narrow portions as described in the above embodiments. Through the above-mentioned method, the production cost can be reduced, the weight of the finished product is reduced due to the reduction of width, it is more lightweight and comfortable when used, the shape of belt with wide and narrow portions is beautiful, and the user experience is improved.
[0085] In some embodiments, the present disclosure further provides an equipment for weaving a double-layer handlebar belt with wide and narrow portions, and the equipment can execute the method having the above-mentioned technical solutions to weave the double-layer handlebar belt with wide and narrow portions as shown in FIG. **1**.
[0086] The above description is merely some embodiments. It should be noted that for one with ordinary skills in the art, improvements can be made without departing from the concept of the present disclosure, but these improvements shall fall into the protection scope of the present disclosure.

Claims

1. A double-layer handlebar belt with wide and narrow portions, comprising: single-layer segments; first double-layer segments alternately connected with the single-layer segments, the first double-layer segment comprises a first layer and a second layer, the first layer is arranged below the second layer, a width of the first layer is greater than that of the second layer; and second double-layer segments alternately connected with the single-layer segments, the second double-layer segment comprises a third layer and a fourth layer, the third layer is arranged below the fourth layer, a width of the third layer is greater than that of the fourth layer, and the fourth layer has a hollow structure, wherein the single-layer segments, the first double-layer segments and the second double-layer segment are woven in one piece.
2. The double-layer handlebar belt with wide and narrow portions according to claim 1, wherein the single-layer segment comprises a first single-layer segment and a second single-layer segment; a first end of the first layer and a first end of the second layer are connected to the first single-layer segment; a second end of the first layer and a second end of the second layer are connected with a first end of the second single-layer segment; and a first end of the third layer and a first end of the fourth layer are connected to a second end of the second single-layer segment.
3. The double-layer handlebar belt with wide and narrow portions of claim 2, wherein the second layer further comprises a third end arranged at a middle of the second layer; the first end and the second end of the second layer are respectively arranged at two ends of the second layer; wherein a width of the first end of the second layer and a width of the second end of the second layer are greater than a width of the third end of the second layer.
4. The double-layer handlebar belt with wide and narrow portions of claim 1, wherein the second layer is arc shaped with respect to the first layer.
5. The double-layer handlebar belt with wide and narrow portions of claim 1, wherein the single-layer segment, the first double-layer segment and the second double-layer segment are made of elastic fabrics or non-elastic fabrics.
6. The double-layer handlebar belt with wide and narrow portions of claim 1, wherein a material shrinkage rate of the first layer is greater than that of the second layer, and a material shrinkage rate of the third layer is greater than that of the fourth layer.
7. A method for weaving a double-layer handlebar belt with wide and narrow portions, comprising: weaving single layer segments; weaving first double layer segments, the first double-layer segment comprises a first layer and a second layer, the first layer is arranged below the second layer, a width of the first layer is greater than that of the second layer; weaving second double layer segments, the second double-layer segment comprises a third layer and a fourth layer, the third layer is arranged below the fourth layer, a width of the third layer is greater than that of the fourth layer, and the fourth layer has a hollow structure, wherein the single layer segments are connected with the double-layer segments alternately, and the single layer segments and the double-layer segments are woven in one piece.
8. The method of claim 7, wherein the operation “weaving first double layer segments” comprises: weaving a first layer and a second layer respectively; wherein when weaving the second layer, controlling a steel buckle to move upwards, and the woven yarn moving upwards based on the steel buckle from a first position of the steel buckle to a second position of the steel buckle, and reducing a conveying capacity of a weft yarn, as such the width of the second layer is reduced from a first width to a second width; controlling the steel buckle to move downwards, the woven yarn moving downwards based on the steel buckle from the second position of the steel buckle to the first position of the steel buckle, and increasing the conveying capacity of the weft yarn, as such the width of the second layer is increased from the second width to the first width; and a width of the first position of the steel buckle is greater than that of the second position of the steel buckle.
9. The method of claim 7, wherein the operation “weaving second double layer segments” comprises: weaving a third layer and a fourth layer respectively; wherein when weaving the fourth

layer, controlling a steel buckle to move upwards, and the woven yarn moving upwards based on the steel buckle from a first position of the steel buckle to a second position of the steel buckle, and reducing a conveying capacity of a weft yarn, as such the width of the second layer is reduced from a first width to a second width; wherein a width of the first position of the steel buckle is greater than that of the second position of the steel buckle.

10. The method of claim 7, wherein the single-layer segment comprises a first single-layer segment and a second single-layer segment; a first end of the first layer and a first end of the second layer are connected to the first single-layer segment; a second end of the first layer and a second end of the second layer are connected with a first end of the second single-layer segment; and a first end of the third layer and a first end of the fourth layer are connected to a second end of the second single-layer segment.

11. The method of claim 10, wherein the second layer further comprises a third end arranged at a middle of the second layer; the first end and the second end of the second layer are respectively arranged at two ends of the second layer; wherein a width of the first end of the second layer and a width of the second end of the second layer are greater than a width of the third end of the second layer.

12. The method of claim 7, wherein the second layer is arc shaped with respect to the first layer.

13. The method of claim 7, wherein the single-layer segment, the first double-layer segment and the second double-layer segment are made of elastic fabrics or non-elastic fabrics.

14. The method of claim 7, wherein a material shrinkage rate of the first layer is greater than that of the second layer, and a material shrinkage rate of the third layer is greater than that of the fourth layer.

15. An equipment for weaving a double-layer handlebar belt with wide and narrow portions, wherein the equipment is configured to execute a method as described in claim 7 and weave the double-layer handlebar belt with wide and narrow portions as described in claim 1.

16. The equipment of claim 15, wherein the operation “weaving first double layer segments” executed by the equipment comprises: weaving a first layer and a second layer respectively; wherein when weaving the second layer, controlling a steel buckle to move upwards, and the woven yarn moving upwards based on the steel buckle from a first position of the steel buckle to a second position of the steel buckle, and reducing a conveying capacity of a weft yarn, as such the width of the second layer is reduced from a first width to a second width; controlling the steel buckle to move downwards, the woven yarn moving downwards based on the steel buckle from the second position of the steel buckle to the first position of the steel buckle, and increasing the conveying capacity of the weft yarn, as such the width of the second layer is increased from the second width to the first width; and a width of the first position of the steel buckle is greater than that of the second position of the steel buckle.

17. The equipment of claim 15, wherein the operation “weaving second double layer segments” executed by the equipment comprises: weaving a third layer and a fourth layer respectively; wherein when weaving the fourth layer, controlling a steel buckle to move upwards, and the woven yarn moving upwards based on the steel buckle from a first position of the steel buckle to a second position of the steel buckle, and reducing a conveying capacity of a weft yarn, as such the width of the second layer is reduced from a first width to a second width; wherein a width of the first position of the steel buckle is greater than that of the second position of the steel buckle.

18. The equipment of claim 15, wherein the single-layer segment comprises a first single-layer segment and a second single-layer segment; a first end of the first layer and a first end of the second layer are connected to the first single-layer segment; a second end of the first layer and a second end of the second layer are connected with a first end of the second single-layer segment; and a first end of the third layer and a first end of the fourth layer are connected to a second end of the second single-layer segment.

19. The equipment of claim 18, wherein the second layer further comprises a third end arranged at a middle of the second layer; the first end and the second end of the second layer are respectively arranged at two ends of the second layer; wherein a width of the first end of the second layer and a width of the second end of the second layer are greater than a width of the third end of the second layer.

20. The equipment of claim 15, wherein the second layer is arc shaped with respect to the first layer.